ASAR TEAM / CURRICULUM VITAE

Abdallah Refaat Allam

Team Leader

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PROFILE

First-year IT student passionate about Robotics and Embedded Systems. Currently working on an autonomous robot using Arduino and ESP32, with hands-on experience in sensors, motor control, and communication systems. Eager to learn and gain practical experience in robotics.

TEAM EXPERIENCE

Team Leader

ASAR Team

ENGINEERING SKILLS

- Embedded Systems & Firmware (C/C++)
- High-Level Scripting & Middleware (Python)
- Version Control & Collaboration (Git/GitHub)
- Embedded Linux & SBC Administration
- Circuit Design & EDA Routing
- Power Electronics & Circuit Protection
- Mechanical Engineering & CAD
- Kinematics & Motion Control
- Sensor Fusion & State Estimation
- Computer Vision & Spatial AI
- SLM Fine tuning & Edge AI deployment

TECHNICAL TOOLKIT

- Autodesk Fusion 360 / SolidWorks
- KiCad EDA (v7.0+) / Proetus
- PlatformIO / VS Code / Arduino IDE 2.x
- Raspberry Pi OS (64-bit) / Ubuntu Server
- ROS 2 (Humble Hawksbill)
- slam_toolbox & ROS 2 Nav2
- OpenCV & ONNX Runtime / TFLite
- FastAPI (Python) / Express & WebSockets
- Gazebo / RViz2 / WebSockets Injection

WORKING STYLE

- Systematic Root-Cause Isolatio
- Critical Thinking & Architecture Trade-Off Analysis
- Safety-First Engineering Mindset (Fail-Safe Design)
- Adaptive Problem Solving & Agile Iteration
- Cross-Domain Systems Integration